

gemini.md — Project Constitution

AI-Powered Asset Management & Maintenance Solution (OCP)

STATUS: BLUEPRINT PHASE — EXPANDED TO 4 MODULES, AWAITING APPROVAL

1. Project Identity

- **Project Name:** OCP Maintenance AI MVP
- **Client:** OCP (Office Cherifien des Phosphates) — Phosphate Mining, Morocco
- **Firm:** Value Strategy Consulting (VSC)
- **Domain:** Industrial Asset Management & Maintenance
- **Industry:** Phosphate Mining (15 plants)
- **Pilot Scope:** 1-2 critical equipment items in 1 plant

2. Discovery Answers (Confirmed 2026-02-20)

#	Question	Answer
1	North Star	Build 4-module OCP MVP Pilot (3 original + Strategy Development)
2	Integrations	SAP PM + Claude API (priority)
3	Source of Truth	Generate phosphate-realistic synthetic data
4	Delivery Payload	Local prototype (Streamlit/Retool), cloud later
5	Behavioral Rules	Safety-first AI (human validates everything)
6	Language	French + English + Arabic (trilingual)
7	Data Fidelity	Phosphate-specific equipment and failure modes

3. Data Schemas

3.1 Equipment Hierarchy (SAP PM Structure)

```
{
  "EquipmentHierarchy": {
    "plant": {
      "plant_id": "string (SAP Plant code, e.g., 'OCP-JFC1')",
      "name": "string",
      "name_fr": "string",
      "name_ar": "string",
      "location": "string (GPS coordinates or site name)"
    },
    "functional_location": {
      "func_loc_id": "string (SAP TPLNR, e.g., 'JFC1-MIN-BRY-01')",
      "description": "string",
      "description_fr": "string",
      "level": "int (1=Plant, 2=Area, 3=System, 4=Sub-system)",
      "parent_func_loc_id": "string | null",
      "plant_id": "string"
    },
    "equipment": {
      "equipment_id": "string (SAP EQUNR, e.g., 'EQ-SAG-001')",
      "tag": "string (Technical TAG, e.g., 'BRY-SAG-ML-001')",
      "description": "string",
      "description_fr": "string",
      "equipment_type": "string (SAG Mill, Conveyor, Crusher, etc.)",
      "manufacturer": "string",
      "model": "string",
      "serial_number": "string",
      "installation_date": "ISO 8601 date",
      "criticality": "enum (AA, A+, A, B, C, D)",
      "func_loc_id": "string",
      "status": "enum (ACTIVE, INACTIVE, DECOMMISSIONED)",
      "weight_kg": "number | null",
      "power_kw": "number | null"
    },
    "component": {
      "component_id": "string (SAP sub-equipment)",
      "description": "string",
      "description_fr": "string",
      "parent_equipment_id": "string",
      "component_type": "string (Motor, Bearing, Liner, Gear, etc.)",
      "manufacturer": "string",
      "part_number": "string"
    }
  }
}
```

3.2 Work Request — Field Capture Input (Raw)

```
{
  "FieldCaptureInput": {
    "capture_id": "UUID",
    "timestamp": "ISO 8601 datetime",
    "technician_id": "string",
    "technician_name": "string",
    "capture_type": "enum (VOICE, TEXT, IMAGE, VOICE+IMAGE)",
    "language_detected": "enum (fr, en, ar)",
    "raw_voice_text": "string | null (transcribed voice input)",
    "raw_text_input": "string | null (typed text input)",
    "images": [
      {
        "image_id": "UUID",
        "file_path": "string",
        "capture_timestamp": "ISO 8601 datetime",
        "gps_coordinates": "string | null"
      }
    ],
    "equipment_tag_manual": "string | null (if technician typed/selected TAG)",
    "location_hint": "string | null (area name spoken or typed)"
  }
}
```

3.3 Structured Work Request — AI-Processed Output

```
{
  "StructuredWorkRequest": {
    "request_id": "UUID",
    "source_capture_id": "UUID (links to FieldCaptureInput)",
    "created_at": "ISO 8601 datetime",
    "status": "enum (DRAFT, PENDING_VALIDATION, VALIDATED, REJECTED, SUBMITTED_TO_SAP)",

    "equipment_identification": {
      "equipment_id": "string (SAP EQUNR, AI-resolved)",
      "equipment_tag": "string (AI-resolved from voice/text/image)",
      "confidence_score": "float (0.0-1.0)",
      "resolution_method": "enum (EXACT_MATCH, FUZZY_MATCH, IMAGE_OCR, MANUAL)"
    },

    "problem_description": {
      "original_text": "string (technician's words)",
      "structured_description": "string (AI-standardized description)",
      "structured_description_fr": "string",
      "failure_mode_detected": "string | null (e.g., 'Vibration excessive')",
      "failure_mode_code": "string | null (links to FMEA if available)",
      "affected_component": "string | null"
    },

    "ai_classification": {
      "work_order_type": "enum (PM01_INSPECTION, PM02_PREVENTIVE, PM03_CORRECTIVE)",
      "priority_suggested": "enum (1_EMERGENCY, 2_URGENT, 3_NORMAL, 4_PLANNED)",
      "priority_justification": "string (AI's reasoning)",
      "estimated_duration_hours": "float",
      "required_specialties": ["string (MECHANICAL, ELECTRICAL, INSTRUMENTATION, etc.)"],
      "safety_flags": ["string (LOCKOUT_TAGOUT, CONFINED_SPACE, HOT_WORK, etc.)"]
    },

    "spare_parts_suggested": [
      {
        "sap_material_code": "string",
        "description": "string",
        "quantity_needed": "int",
        "availability_status": "enum (IN_STOCK, LOW_STOCK, OUT_OF_STOCK, UNKNOWN)",
        "warehouse_location": "string | null",
        "lead_time_days": "int | null"
      }
    ],

    "image_analysis": {
      "anomalies_detected": ["string"],
      "component_identified": "string | null",
      "severity_visual": "enum (LOW, MEDIUM, HIGH, CRITICAL) | null"
    },

    "validation": {
      "validated_by": "string | null (planner ID)",
      "validated_at": "ISO 8601 datetime | null",
      "modifications_made": ["string (list of fields changed by planner)"],
      "final_priority": "enum (1_EMERGENCY, 2_URGENT, 3_NORMAL, 4_PLANNED) | null"
    }
  }
}
```

```
}  
}
```

3.4 Planner Recommendation (AI Planner Assistant Output)

```
{
  "PlannerRecommendation": {
    "recommendation_id": "UUID",
    "work_request_id": "UUID (links to StructuredWorkRequest)",
    "generated_at": "ISO 8601 datetime",

    "resource_analysis": {
      "workforce_available": [
        {
          "specialty": "string",
          "technicians_available": "int",
          "next_available_slot": "ISO 8601 datetime"
        }
      ],
      "materials_status": {
        "all_available": "boolean",
        "missing_items": [
          {
            "material_code": "string",
            "description": "string",
            "estimated_arrival": "ISO 8601 date | null",
            "alternative_available": "boolean",
            "alternative_code": "string | null"
          }
        ]
      }
    },
    "shutdown_window": {
      "next_available": "ISO 8601 datetime | null",
      "type": "enum (MINOR_8H, MAJOR_20H_PLUS) | null",
      "duration_hours": "float | null"
    },
    "production_impact": {
      "estimated_downtime_hours": "float",
      "production_loss_tons": "float | null",
      "cost_estimate_usd": "float | null"
    }
  },
  "scheduling_suggestion": {
    "recommended_date": "ISO 8601 date",
    "recommended_shift": "enum (MORNING, AFTERNOON, NIGHT)",
    "reasoning": "string",
    "conflicts": ["string (list of scheduling conflicts detected)"],
    "groupable_with": ["string (work request IDs that could be grouped)"]
  },
  "risk_assessment": {
    "risk_level": "enum (LOW, MEDIUM, HIGH, CRITICAL)",
    "risk_factors": ["string"],
    "recommendation": "string"
  },
  "planner_action_required": "enum (APPROVE, MODIFY, ESCALATE, DEFER)",
  "ai_confidence": "float (0.0-1.0)"
}
```

3.5 Backlog Item

```
{
  "BacklogItem": {
    "backlog_id": "UUID",
    "work_request_id": "UUID",
    "equipment_id": "string",
    "equipment_tag": "string",
    "priority": "enum (1_EMERGENCY, 2_URGENT, 3_NORMAL, 4_PLANNED)",
    "work_order_type": "enum (PM01, PM02, PM03)",
    "created_date": "ISO 8601 date",
    "age_days": "int",
    "status": "enum (AWAITING_MATERIALS, AWAITING_SHUTDOWN, AWAITING_RESOURCES, AWAITING_APPROVAL, SC",
    "blocking_reason": "string | null",
    "estimated_duration_hours": "float",
    "required_specialties": ["string"],
    "materials_ready": "boolean",
    "shutdown_required": "boolean",
    "groupable": "boolean",
    "group_id": "string | null (work package group)"
  }
}
```

3.6 Optimized Backlog Output

```
{
  "OptimizedBacklog": {
    "optimization_id": "UUID",
    "generated_at": "ISO 8601 datetime",
    "period": {
      "start_date": "ISO 8601 date",
      "end_date": "ISO 8601 date"
    },
    "summary": {
      "total_backlog_items": "int",
      "items_schedulable_now": "int",
      "items_blocked": "int",
      "estimated_total_hours": "float"
    },
    "stratification": {
      "by_reason": {
        "awaiting_materials": "int",
        "awaiting_shutdown": "int",
        "awaiting_resources": "int",
        "awaiting_approval": "int",
        "schedulable": "int"
      },
      "by_priority": {
        "emergency": "int",
        "urgent": "int",
        "normal": "int",
        "planned": "int"
      },
      "by_equipment_criticality": {
        "AA": "int",
        "A_plus": "int",
        "A": "int",
        "B": "int",
        "C": "int",
        "D": "int"
      }
    },
    "work_packages": [
      {
        "package_id": "string",
        "name": "string",
        "grouped_items": ["string (backlog_ids)"],
        "reason_for_grouping": "string (same equipment, same area, same shutdown, etc.)",
        "scheduled_date": "ISO 8601 date",
        "scheduled_shift": "enum (MORNING, AFTERNOON, NIGHT)",
        "total_duration_hours": "float",
        "assigned_team": ["string (specialty + count)"],
        "materials_status": "enum (READY, PARTIAL, NOT_READY)"
      }
    ],
    "schedule_proposal": [
      {
```



```

        "date": "ISO 8601 date",
        "shift": "enum (MORNING, AFTERNOON, NIGHT)",
        "work_packages": ["string (package_ids)"],
        "total_hours": "float",
        "utilization_percent": "float"
    }
],

"alerts": [
    {
        "type": "enum (OVERDUE, MATERIAL_DELAY, RESOURCE_CONFLICT, PRIORITY_ESCALATION)",
        "message": "string",
        "affected_items": ["string (backlog_ids)"]
    }
]
}
}

```

3.7 Work Order History (SAP PM Import Format)

```

{
  "WorkOrderHistory": {
    "work_order_id": "string (SAP AUFNR)",
    "order_type": "enum (PM01, PM02, PM03)",
    "equipment_id": "string (SAP EQUNR)",
    "equipment_tag": "string",
    "func_loc_id": "string (SAP TPLNR)",
    "description": "string",
    "description_fr": "string",
    "priority": "enum (1, 2, 3, 4)",
    "status": "enum (CREATED, RELEASED, IN_PROGRESS, COMPLETED, CLOSED, CANCELLED)",
    "created_date": "ISO 8601 date",
    "planned_start": "ISO 8601 date",
    "planned_end": "ISO 8601 date",
    "actual_start": "ISO 8601 date | null",
    "actual_end": "ISO 8601 date | null",
    "actual_duration_hours": "float | null",
    "man_hours": "float | null",
    "problem_description": "string",
    "cause_description": "string | null",
    "solution_description": "string | null",
    "materials_consumed": [
      {
        "material_code": "string",
        "description": "string",
        "quantity": "float",
        "unit": "string"
      }
    ],
    "assigned_team": "string",
    "postponement_reason": "string | null",
    "cost_total": "float | null"
  }
}

```

3.8 Spare Parts / BOM

```
{
  "SparePart": {
    "material_code": "string (SAP MATNR)",
    "description": "string",
    "description_fr": "string",
    "material_group": "string",
    "applicable_equipment": ["string (equipment_ids)"],
    "manufacturer": "string",
    "manufacturer_part_number": "string",
    "unit_of_measure": "string",
    "criticality": "enum (CRITICAL, IMPORTANT, STANDARD)",
    "lead_time_days": "int",
    "supplier": "string",
    "unit_cost_usd": "float"
  }
}
```

```
{
  "InventoryItem": {
    "material_code": "string",
    "warehouse_id": "string",
    "warehouse_location": "string",
    "quantity_on_hand": "float",
    "quantity_reserved": "float",
    "quantity_available": "float",
    "min_stock": "float",
    "safety_stock": "float",
    "reorder_point": "float",
    "last_movement_date": "ISO 8601 date"
  }
}
```

3.9 Maintenance Plan (SAP PM Preventive)

```
{
  "MaintenancePlan": {
    "plan_id": "string (SAP maintenance plan number)",
    "description": "string",
    "description_fr": "string",
    "equipment_id": "string",
    "equipment_tag": "string",
    "strategy": "enum (TIME_BASED, CONDITION_BASED, PREDICTIVE)",
    "frequency_days": "int",
    "frequency_unit": "enum (DAYS, WEEKS, MONTHS, HOURS_RUN)",
    "task_list": [
      {
        "task_id": "string",
        "description": "string",
        "description_fr": "string",
        "duration_hours": "float",
        "specialty_required": "string",
        "spare_parts": ["string (material_codes)"]
      }
    ],
    "last_execution_date": "ISO 8601 date",
    "next_execution_date": "ISO 8601 date",
    "status": "enum (ACTIVE, SUSPENDED, DELETED)"
  }
}
```

4. Behavioral Rules (CONFIRMED)

4.1 Safety-First AI (MANDATORY)

1. **AI NEVER auto-submits to SAP.** All AI outputs require explicit human validation.
2. **Planner always has final say.** AI suggests, human decides.
3. **Conservative priority:** Better to over-flag risk than under-flag.
4. **No autonomous actions:** Draft → Human Review → Approve → Execute.
5. **Audit trail:** Every AI decision must be logged with reasoning and confidence score.

4.2 Language Rules

1. **Trilingual support:** French (primary), English (technical), Arabic (field input).
2. **Field input:** Accept voice/text in any of the 3 languages. Auto-detect language.
3. **AI output:** Generate structured descriptions in French + English.
4. **UI:** Switchable language preference per user.
5. **Technical terms:** Always preserve SAP codes, TAGs, and part numbers as-is.

4.3 Data Rules

1. **Synthetic data first:** All development uses phosphate-realistic synthetic data.
2. **SAP format compliance:** All data follows SAP PM field conventions.
3. **PII redaction:** Technician names redacted before LLM processing.
4. **Immutable records:** Work requests, once validated, cannot be modified (only new versions).

4.4 Failure Mode Rules (MANDATORY)

1. **ALL failure modes MUST use a valid Mechanism + Cause combination from the authoritative lookup table:** `MAINTENANCE STRATEGY.../Failure Modes (Mechanism + Cause).xlsx` (SRC-09 in DOCUMENT_INDEX).
2. **There are exactly 72 valid combinations** across 18 mechanisms. No other combinations are permitted.
3. **Any agent, assistant, or code** that generates, suggests, or validates failure modes MUST validate against this table.
4. **The 18 valid mechanisms are:** Arcs, Blocks, Breaks/Fracture/Separates, Corrodes, Cracks, Degrades, Distorts, Drifts, Expires, Immobilised (binds/jams), Looses Preload, Open-Circuit, Overheats/Melts, Severs (cut/tear/hole), Short-Circuits, Thermally Overloads, Washes Off, Wears.
5. **Each mechanism has specific valid causes** — e.g., "Wears" can ONLY be caused by: Breakdown of lubrication, Entrained air, Excessive fluid velocity, Impact/shock loading, Low pressure, Lubricant contamination (particles), Mechanical overload, Metal to metal contact, Relative movement between contacting surfaces.
6. **For reliability investigations**, the table also maps "Other Mechanisms" and "Other Causes" grouped under each FM — use these for root cause analysis categorization.

4.5 Do-Not Rules

1. **DO NOT** auto-escalate priorities without planner confirmation.
2. **DO NOT** delete or modify historical work order data.
3. **DO NOT** send raw field input (voice/images) to external APIs without PII check.
4. **DO NOT** schedule work during production windows without explicit override.
5. **DO NOT** suggest spare parts substitutions without flagging to planner.
6. **DO NOT** create failure modes with Mechanism+Cause combinations not present in SRC-09 (Failure Modes lookup table).

5. Architectural Invariants

5.1 A.N.T. 3-Layer Architecture

- **Layer 1 (Architecture):** All SOPs in `architecture/` as Markdown

- **Layer 2 (Navigation):** Routing/decision logic — never performs complex tasks directly
- **Layer 3 (Tools):** Deterministic Python scripts in `tools/` — atomic, testable

5.2 Core Principles

- LLMs are probabilistic; business logic MUST be deterministic
- Update SOP before updating code (Golden Rule)
- Environment variables in `.env` only
- Intermediate files in `.tmp/` only
- Self-Annealing: Analyze -> Patch -> Test -> Update Architecture on every failure

5.3 Security Invariants

- PII redaction before any LLM processing (Microsoft Presidio or equivalent)
- Row-Level Security (RLS) on all database tables
- Immutable audit trails for all work orders and AI decisions
- SAP integration: read-only first, write after sandbox validation

5.4 MVP Technology Stack (Confirmed)

Layer	Technology	Purpose
AI/NLP	Claude API (Sonnet 4)	Work request processing, classification, recommendations
Database	PostgreSQL (local for prototype)	Equipment, work orders, backlog, spare parts
Backend	Python (FastAPI)	API services, business logic
UI (Prototype)	Streamlit	Planner dashboard, backlog visualization
Data Generation	Python (Faker + domain logic)	Phosphate-realistic synthetic SAP PM data
SAP Integration	Mock API (JSON files)	Simulated SAP PM endpoints for MVP
File Storage	Local filesystem (.tmp/)	Voice transcripts, images, intermediate data

6. Integration Registry

Service	Status	API Key Location	MVP Approach
SAP PM	MOCK	.env	JSON mock files simulating IW38/IW39/IP10
Claude API	TO VERIFY	.env (ANTHROPIC_API_KEY)	Direct API for NLP/classification
PostgreSQL	TO SET UP	.env (DATABASE_URL)	Local instance for prototype
Deepgram	DEFERRED	-	Voice transcription (Phase 2)
PI System	DEFERRED	-	Real-time data (Phase 2)

7. Multi-Agent Architecture (Claude Agent SDK)

7.1 Architecture Overview

The R8 methodology is executed by **AI agents**, not by humans using software. The human invokes the work, agents perform the analysis (FMEA, RCM, task generation, work packaging, SAP upload), and the human approves at 4 milestone gates.

```
Human: "Develop strategy for SAG Mill 001"
|
▼
Orchestrator Agent (sonnet) — coordinates workflow, 4 milestones
|
├── Reliability Agent (opus)
│   29 tools: criticality, RCM, FM validation, Weibull, KPI, health
|
├── Planning Agent (sonnet)
│   20 tools: backlog grouping, SAP export, work instructions, CAPA
|
└── Spare Parts Agent (haiku)
    3 tools: material suggestion, validation, equipment resolution
```

Existing code (UNCHANGED): 15 engines, 3 validators, 1 generator, 31 Pydantic models

New code (ADDITIVE): `agents/` directory — tool wrappers, agent definitions, orchestration

7.2 Directory Structure

```
agents/
├── __init__.py
├── run.py                                # CLI entry point
├── tool_wrappers/
│   ├── registry.py                     # @tool() decorator + TOOL_REGISTRY
│   ├── server.py                       # Unified server + AGENT_TOOL_MAP
│   ├── criticality_tools.py            # 4 tools
│   ├── rcm_tools.py                   # 2 tools
│   ├── sap_tools.py                   # 3 tools
│   ├── priority_tools.py              # 2 tools
│   ├── backlog_tools.py               # 5 tools
│   ├── equipment_tools.py             # 1 tool
│   ├── material_tools.py              # 2 tools
│   ├── work_instruction_tools.py       # 2 tools
│   ├── state_machine_tools.py         # 3 tools
│   ├── validation_tools.py            # 14 tools
│   ├── fm_lookup_tools.py             # 4 tools
│   ├── health_tools.py                # 2 tools
│   ├── kpi_tools.py                   # 5 tools
│   ├── weibull_tools.py               # 3 tools
│   ├── variance_tools.py              # 3 tools
│   ├── capa_tools.py                  # 7 tools
│   └── management_review_tools.py      # 1 tool
├── definitions/
│   ├── base.py                        # Agent class + agentic tool-use loop
│   ├── orchestrator.py                # OrchestratorAgent + delegation
│   ├── reliability.py                  # Reliability Engineer (opus)
│   ├── planning.py                    # Planning Specialist (sonnet)
│   ├── spare_parts.py                 # Spare Parts Specialist (haiku)
│   └── prompts/
│       ├── orchestrator_prompt.md
│       ├── reliability_prompt.md
│       ├── planning_prompt.md
│       └── spare_parts_prompt.md
└── orchestration/
    ├── session_state.py                # SessionState accumulator
    ├── milestones.py                   # MilestoneGate + 4 definitions
    └── workflow.py                     # StrategyWorkflow + human gates
```

7.3 Four Milestone Gates

#	Milestone	Agents	Human Reviews
1	Hierarchy Decomposition	Reliability	Equipment breakdown (6 levels), criticality
2	FMEA Completion	Reliability	Failure modes (72-combo validated), RCM paths
3	Strategy + Tasks + Resources	Reliability + Planning + Spare Parts	Tasks, WPs, materials, work instructions
4	SAP Upload Package	Planning	Maintenance Item + Task List + Work Plan (DRAFT)

Gate flow: `PENDING` → `IN_PROGRESS` → `PRESENTED` → `APPROVED` / `MODIFIED` / `REJECTED`

At each gate the Orchestrator runs `run_full_validation` (40+ quality rules), presents results with error/warning/info counts, and waits for human action.

7.4 Agent-to-Tool Mapping (62 tools total)

Agent	Model	# Tools	Key Tool Categories
Orchestrator	sonnet	6	<code>run_full_validation</code> , <code>evaluate_confidence</code> , <code>validate_state_transition</code>
Reliability	opus	29	criticality, RCM, FM 72-combo, priority, health, KPI, Weibull, variance
Planning	sonnet	20	backlog grouping, SAP export, work instructions, CAPA, naming validation
Spare Parts	haiku	3	<code>suggest_materials</code> , <code>validate_task_materials</code> , <code>resolve_equipment</code>

7.5 Safety-First Rules (Agent Layer)

1. **NEVER skip validation** — `run_full_validation` before every milestone gate
2. **NEVER auto-submit to SAP** — all outputs are DRAFT until human approves
3. **NEVER proceed past a gate without human approval**
4. **NEVER create a failure mode without 72-combo validation** (`validate_fm_combination`)
5. **Flag low-confidence items** — `evaluate_confidence` for all AI-generated data
6. **T-16 rule**: REPLACE tasks MUST have materials assigned

7.6 CLI Usage

```
python -m agents.run "SAG Mill 001" --plant OCP-JFC
python -m agents.run "Ball Mill BM-201" --plant OCP-BEN --output session.json
```

8. Module 4: Maintenance Strategy Development — Discovery Answers

#	Question	Answer
8	Alcance	Flujo completo: Jerarquía → Criticidad → FMEA → Estrategia → Tareas → Work Packages → SAP Upload
9	Librerías	Con IA: Component + Equipment Libraries con sugerencia automática de descomposición
10	Rol de IA	Auto-completador: IA genera draft completo, ingeniero revisa y ajusta
11	Materiales	Integrado: catálogo de materiales dentro del módulo de estrategia

9. Data Schemas — Module 4: Strategy Development

9.1 Component Library Item (Generic Reusable)

```
{
  "ComponentLibraryItem": {
    "component_lib_id": "UUID",
    "name": "string (e.g., 'Centrifugal Pump, Slurry Service')",
    "code": "string (unique library code)",
    "component_category": "enum (MECHANICAL, ELECTRICAL, INSTRUMENTATION, STRUCTURAL, HYDRAULIC, PNEUMATIC)",
    "description": "string",
    "description_fr": "string",
    "typical_manufacturers": ["string"],
    "functions": ["Function"],
    "failure_modes": ["FailureMode"],
    "default_tasks": ["MaintenanceTask"],
    "tags": ["string (searchable tags: 'pump', 'slurry', 'centrifugal')"],
    "source": "enum (R8_LIBRARY, OEM, CUSTOM, AI_GENERATED)",
    "version": "int",
    "locked": "boolean",
    "created_at": "ISO 8601 datetime",
    "updated_at": "ISO 8601 datetime"
  }
}
```

9.2 Equipment Library Item (Composed from Components)

```
{
  "EquipmentLibraryItem": {
    "equipment_lib_id": "UUID",
    "name": "string (Make - Model - Context, e.g., 'Warman 750 VK - SHD Pump - Slurry Service')",
    "code": "string (unique library code)",
    "equipment_category": "enum (PUMP, CONVEYOR, CRUSHER, MILL, FLOTATION_CELL, THICKENER, DRYER, KILN)",
    "make": "string",
    "model": "string",
    "operational_context": "string (e.g., 'Slurry Service', 'Water Service', 'Acid Service')",
    "description": "string",
    "description_fr": "string",
    "sub_assemblies": [
      {
        "sub_assembly_id": "UUID",
        "name": "string (e.g., 'Mechanical System', 'Electrical System', 'Drive System')",
        "order": "int",
        "components": [
          {
            "instance_id": "UUID",
            "component_lib_ref": "UUID (→ ComponentLibraryItem)",
            "instance_name": "string (contextualized name, e.g., 'Drive End Bearing')",
            "quantity": "int (default 1)",
            "is_maintainable_item": "boolean"
          }
        ]
      }
    ],
    "source": "enum (R8_LIBRARY, OEM, CUSTOM, AI_GENERATED)",
    "version": "int",
    "locked": "boolean",
    "created_at": "ISO 8601 datetime",
    "updated_at": "ISO 8601 datetime"
  }
}
```

9.3 Plant Hierarchy Node (Site-Specific)

```
{
  "PlantHierarchyNode": {
    "node_id": "UUID",
    "node_type": "enum (PLANT, AREA, SYSTEM, EQUIPMENT, SUB_ASSEMBLY, MAINTAINABLE_ITEM)",
    "name": "string",
    "name_fr": "string",
    "code": "string (SAP functional location or equipment code)",
    "parent_node_id": "UUID | null",
    "level": "int (1=Plant, 2=Area, 3=System, 4=Equipment, 5=Sub-Assembly, 6=MI)",
    "equipment_lib_ref": "UUID | null (→ EquipmentLibraryItem, for equipment nodes)",
    "component_lib_ref": "UUID | null (→ ComponentLibraryItem, for MI nodes)",
    "sap_func_loc": "string | null (SAP TPLNR)",
    "sap_equipment_nr": "string | null (SAP EQUNR)",
    "tag": "string | null (technical TAG)",
    "criticality": "CriticalityAssessment | null",
    "status": "enum (ACTIVE, INACTIVE, DECOMMISSIONED)",
    "order": "int (display order among siblings)",
    "metadata": {
      "manufacturer": "string | null",
      "model": "string | null",
      "serial_number": "string | null",
      "installation_date": "ISO 8601 date | null",
      "power_kw": "number | null",
      "weight_kg": "number | null",
      "operational_hours": "number | null"
    }
  }
}
```

9.4 Criticality Assessment

```
{
  "CriticalityAssessment": {
    "assessment_id": "UUID",
    "node_id": "UUID (→ PlantHierarchyNode)",
    "assessed_at": "ISO 8601 datetime",
    "assessed_by": "string",
    "method": "enum (FULL_MATRIX, SIMPLIFIED)",
    "criteria_scores": [
      {
        "category": "enum (SAFETY, HEALTH, ENVIRONMENT, PRODUCTION, OPERATING_COST, CAPITAL_COST, SCH",
        "consequence_level": "int (1-5)",
        "comments": "string | null"
      }
    ],
    "probability": "int (1-5)",
    "overall_score": "float (auto-calculated)",
    "risk_class": "enum (I_LOW, II_MEDIUM, III_HIGH, IV_CRITICAL)",
    "ai_suggested_class": "enum (I_LOW, II_MEDIUM, III_HIGH, IV_CRITICAL) | null",
    "ai_justification": "string | null",
    "status": "enum (DRAFT, REVIEWED, APPROVED)"
  }
}
```

9.5 Function

```
{
  "Function": {
    "function_id": "UUID",
    "node_id": "UUID (→ PlantHierarchyNode or ComponentLibraryItem)",
    "function_type": "enum (PRIMARY, SECONDARY, PROTECTIVE)",
    "description": "string (Verb + Noun + Performance Standard)",
    "description_fr": "string",
    "performance_standard": "string | null",
    "functional_failures": ["FunctionalFailure"],
    "ai_generated": "boolean",
    "status": "enum (DRAFT, REVIEWED, APPROVED)"
  }
}
```

9.6 Functional Failure

```
{
  "FunctionalFailure": {
    "failure_id": "UUID",
    "function_id": "UUID (→ Function)",
    "failure_type": "enum (TOTAL, PARTIAL)",
    "description": "string",
    "description_fr": "string",
    "failure_modes": ["FailureMode"]
  }
}
```

9.7 Failure Mode

```
{
  "FailureMode": {
    "failure_mode_id": "UUID",
    "functional_failure_id": "UUID (→ FunctionalFailure)",
    "status": "enum (RECOMMENDED, REDUNDANT)",
    "what": "string (sub-component, capital letter, singular)",
    "mechanism": "enum – MUST be one of the 18 mechanisms from SRC-09: ARCS, BLOCKS, BREAKS_FRACTURE_...
    "cause": "string – MUST be a valid cause for the selected mechanism per SRC-09 lookup table (72 v...
    "failure_pattern": "enum (A_BATHTUB, B_AGE, C_FATIGUE, D_STRESS, E_RANDOM, F_EARLY_LIFE) | null",
    "failure_consequence": "enum (HIDDEN_SAFETY, HIDDEN_NONSAFETY, EVIDENT_SAFETY, EVIDENT_ENVIRONMEN...
    "is_hidden": "boolean",
    "failure_effect": {
      "evidence": "string (what operator sees/hears/feels)",
      "safety_threat": "string | null",
      "environmental_threat": "string | null",
      "production_impact": "string | null",
      "physical_damage": "string | null",
      "repair_description": "string | null",
      "estimated_downtime_hours": "float | null"
    },
    "strategy_type": "enum (CONDITION_BASED, FIXED_TIME, RUN_TO_FAILURE, FAULT_FINDING, REDESIGN, OEM...
    "primary_task": "MaintenanceTask",
    "secondary_task": "MaintenanceTask | null (required for CB/FFI)",
    "ai_generated": "boolean",
    "ai_confidence": "float | null",
    "existing_task_source": "string | null (R8 Library, MSO, OEM, Workshop)",
    "justification_category": "enum (MODIFIED, ELIMINATED, FREQUENCY_CHANGE, TACTIC_CHANGE, MAINTAIN...
  }
}
```

9.8 Maintenance Task

```
{
  "MaintenanceTask": {
    "task_id": "UUID",
    "name": "string (standardized: 'Inspect [what] for [evidence]', max 72 chars)",
    "name_fr": "string",
    "task_type": "enum (INSPECT, CHECK, TEST, LUBRICATE, CLEAN, REPLACE, REPAIR, CALIBRATE)",
    "is_secondary": "boolean",
    "acceptable_limits": "string | null (required for CB/FFI)",
    "conditional_comments": "string | null",
    "consequences": "string (what happens if not performed)",
    "justification": "string | null",
    "constraint": "enum (ONLINE, OFFLINE, TEST_MODE)",
    "access_time_hours": "float (0 for online)",
    "frequency_value": "float",
    "frequency_unit": "enum (HOURS, DAYS, WEEKS, MONTHS, YEARS, OPERATING_HOURS, TONNES, CYCLES)",
    "origin": "string | null (OEM, Statutory, Library, Workshop)",
    "budget_type": "enum (REPAIR, REPLACE) | null (secondary tasks only)",
    "budgeted_life": "float | null (secondary tasks only)",
    "labour_resources": [
      {
        "specialty": "enum (FITTER, ELECTRICIAN, INSTRUMENTIST, OPERATOR, COMMON_SPECIALIST, LUBRICAT",
        "quantity": "int",
        "hours_per_person": "float",
        "hourly_rate": "float | null"
      }
    ],
    "material_resources": [
      {
        "material_code": "string | null (SAP MATNR)",
        "description": "string",
        "part_number": "string | null",
        "stock_code": "string | null",
        "quantity": "float",
        "unit_of_measure": "enum (EA, L, KG, M)",
        "unit_price": "float | null"
      }
    ],
    "tools": ["string"],
    "special_equipment": ["string (crane, scaffold, etc.)"],
    "ai_generated": "boolean",
    "ai_confidence": "float | null",
    "status": "enum (DRAFT, REVIEWED, APPROVED)"
  }
}
```

9.9 Work Package

```
{
  "WorkPackage": {
    "work_package_id": "UUID",
    "name": "string (max 40 chars, ALL CAPS, format: [FREQ] [ASSET] [TRADE] [TYPE] [CONSTRAINT])",
    "code": "string (unique identifier)",
    "node_id": "UUID (→ PlantHierarchyNode, equipment level)",
    "frequency_value": "float",
    "frequency_unit": "enum (HOURS, DAYS, WEEKS, MONTHS, YEARS, OPERATING_HOURS, TONNES, CYCLES)",
    "constraint": "enum (ONLINE, OFFLINE)",
    "access_time_hours": "float",
    "work_package_type": "enum (STANDALONE, SUPPRESSIVE, SEQUENTIAL)",
    "job_preparation": "string | null (pre-task notes)",
    "post_shutdown": "string | null (post-task notes)",
    "allocated_tasks": [
      {
        "task_id": "UUID (→ MaintenanceTask)",
        "order": "int (execution sequence)",
        "operation_number": "int (SAP: 10, 20, 30...)"
      }
    ],
    "labour_summary": {
      "total_hours": "float",
      "by_specialty": [{"specialty": "string", "hours": "float", "people": "int"}]
    },
    "material_summary": [
      {"material_code": "string", "description": "string", "quantity": "float"}
    ],
    "sap_upload": {
      "maintenance_item_ref": "string | null",
      "task_list_ref": "string | null",
      "work_plan_ref": "string | null"
    },
    "status": "enum (DRAFT, REVIEWED, APPROVED, UPLOADED_TO_SAP)"
  }
}
```

9.10 SAP Upload Package (Generated Output)

```
{
  "SAPUploadPackage": {
    "package_id": "UUID",
    "generated_at": "ISO 8601 datetime",
    "plant_code": "string (SAP IWERK)",
    "maintenance_plan": {
      "plan_id": "string",
      "description": "string",
      "category": "string (PM)",
      "cycle_value": "int",
      "cycle_unit": "string (DAY/WK/MON/YR)",
      "call_horizon_pct": "int",
      "scheduling_period": "int",
      "scheduling_unit": "string"
    },
    "maintenance_items": [
      {
        "item_ref": "string ($MI1, $MI2...)",
        "description": "string",
        "order_type": "string (PM03)",
        "func_loc": "string (SAP TPLNR)",
        "main_work_center": "string",
        "planner_group": "int",
        "task_list_ref": "string ($TL1, $TL2...)",
        "priority": "string"
      }
    ],
    "task_lists": [
      {
        "list_ref": "string ($TL1)",
        "description": "string",
        "func_loc": "string",
        "system_condition": "int (1=running, 3=stopped)",
        "operations": [
          {
            "operation_number": "int (10, 20, 30...)",
            "work_centre": "string",
            "control_key": "string (PMIN)",
            "short_text": "string (max 72 chars)",
            "duration_hours": "float",
            "unit": "string (H)",
            "num_workers": "int"
          }
        ]
      }
    ],
    "status": "enum (GENERATED, REVIEWED, APPROVED, UPLOADED)"
  }
}
```


10. GECAMIN Extensions — Strategic Recommendation Implementations

10.1 Asset Health Score (REF-12 Rec 4, GECAMIN S6: SQM validated)

```
{
  "AssetHealthScore": {
    "score_id": "UUID",
    "node_id": "string (→ PlantHierarchyNode)",
    "plant_id": "string",
    "equipment_tag": "string",
    "calculated_at": "ISO 8601 datetime",
    "dimensions": [
      {
        "dimension": "enum (CRITICALITY, BACKLOG_PRESSURE, STRATEGY_COVERAGE, CONDITION_STATUS, EXECUTION_STATUS)",
        "score": "float (0-100)",
        "weight": "float (0-1, default 0.2)",
        "raw_value": "float | null",
        "details": "string"
      }
    ],
    "composite_score": "float (0-100, auto-calculated)",
    "health_class": "string (HEALTHY ≥75, AT_RISK ≥50, CRITICAL <50, UNKNOWN)",
    "trend": "enum (IMPROVING, STABLE, DEGRADING)",
    "recommendations": ["string"]
  }
}
```

10.2 KPI Metrics (REF-12 Rec 8, ISO 55002 §9.1)

```
{
  "KPIMetrics": {
    "metrics_id": "UUID",
    "plant_id": "string",
    "equipment_id": "string | null",
    "period_start": "ISO 8601 date",
    "period_end": "ISO 8601 date",
    "mtbf_days": "float | null (Mean Time Between Failures)",
    "mttr_hours": "float | null (Mean Time To Repair)",
    "availability_pct": "float | null (0-100%)",
    "oeo_pct": "float | null (Overall Equipment Effectiveness, 0-100%)",
    "schedule_compliance_pct": "float | null (0-100%)",
    "pm_compliance_pct": "float | null (Preventive Maintenance compliance, 0-100%)",
    "reactive_ratio_pct": "float | null (corrective / total WOs, 0-100%)",
    "total_work_orders": "int",
    "corrective_wo_count": "int",
    "preventive_wo_count": "int"
  }
}
```

10.3 Weibull Failure Prediction (REF-12 Rec 6, GECAMIN S3/S7)

```
{
  "FailurePrediction": {
    "prediction_id": "UUID",
    "equipment_id": "string",
    "equipment_tag": "string",
    "predicted_at": "ISO 8601 datetime",
    "weibull_params": {
      "beta": "float (>0, shape: >1=wear-out, <1=early-life, =1=random)",
      "eta": "float (>0, characteristic life in days)",
      "gamma": "float (≥0, failure-free period)",
      "r_squared": "float (0-1, goodness of fit)",
      "sample_size": "int"
    },
    "current_age_days": "float",
    "reliability_current": "float (0-1, R(t) at current age)",
    "predicted_failure_window_days": "float (days until predicted failure)",
    "confidence_level": "float (0.5-0.99)",
    "risk_score": "float (0-100)",
    "failure_pattern": "enum (A_BATHTUB, B_AGE, C_FATIGUE, D_STRESS, E_RANDOM, F_EARLY_LIFE)",
    "recommendation": "string",
    "status": "enum (DRAFT, REVIEWED, APPROVED) – ALWAYS starts as DRAFT (safety-first)"
  }
}
```

10.4 Plant Variance Alert (REF-12 Rec 7)

```
{
  "PlantVarianceAlert": {
    "alert_id": "UUID",
    "plant_id": "string",
    "plant_name": "string",
    "metric_name": "string",
    "plant_value": "float",
    "portfolio_mean": "float",
    "portfolio_std": "float",
    "z_score": "float",
    "variance_level": "enum (NORMAL, WARNING ≥2σ, CRITICAL ≥3σ)",
    "detected_at": "ISO 8601 datetime",
    "message": "string"
  }
}
```

10.5 CAPA Item (REF-12 Rec 8, ISO 55002 §10.2)

```
{
  "CAPAItem": {
    "capa_id": "UUID",
    "capa_type": "enum (CORRECTIVE, PREVENTIVE)",
    "title": "string",
    "description": "string",
    "plant_id": "string",
    "equipment_id": "string | null",
    "source": "string (audit, incident, variance, review)",
    "root_cause": "string | null",
    "current_phase": "enum (PLAN, DO, CHECK, ACT)",
    "status": "enum (OPEN, IN_PROGRESS, CLOSED, VERIFIED)",
    "assigned_to": "string",
    "target_date": "ISO 8601 date | null",
    "actions_planned": ["string"],
    "actions_completed": ["string"],
    "effectiveness_verified": "boolean",
    "lessons_learned": "string | null"
  }
}
```

10.6 Management Review Summary (REF-12 Rec 8, ISO 55002 §9.3)

```
{
  "ManagementReviewSummary": {
    "review_id": "UUID",
    "plant_id": "string",
    "period_start": "ISO 8601 date",
    "period_end": "ISO 8601 date",
    "kpi_summary": "KPIMetrics | null",
    "health_scores": ["AssetHealthScore"],
    "avg_health_score": "float (0-100)",
    "variance_alerts": ["PlantVarianceAlert"],
    "open_capas": "int",
    "overdue_capas": "int",
    "health_trend": "enum (IMPROVING, STABLE, DEGRADING)",
    "kpi_trends": {"metric_name": "IMPROVING | STABLE | DEGRADING"},
    "key_findings": ["string"],
    "recommended_actions": ["string"]
  }
}
```

11. Neuro-Architecture Extensions — Behavioral Science Models

11.1 Expert Card (Pillar 2: TMS + ISO 55002 §4.2)

```
{
  "ExpertCard": {
    "expert_id": "UUID",
    "user_id": "string",
    "name": "string",
    "role": "enum (MAINTENANCE_MANAGER, RELIABILITY_ENGINEER, PLANNER, TECHNICIAN, OPERATOR, PLANT_MA",
    "plant_id": "string",
    "domains": ["enum (RELIABILITY, MECHANICAL, ELECTRICAL, INSTRUMENTATION, PROCESS, SAFETY, PLANNING)",
    "equipment_expertise": ["string (equipment TAGs)"],
    "certifications": ["string"],
    "years_experience": "int",
    "resolution_count": "int",
    "languages": ["enum (fr, en, ar)"]
  }
}
```

11.2 Ipsative Feedback (Pillar 5: Psychological Safety)

```
{
  "IpsativeFeedback": {
    "feedback_id": "UUID",
    "user_id": "string",
    "metric_name": "string",
    "current_value": "float",
    "previous_value": "float",
    "improvement_pct": "float (auto-calculated)",
    "period": "string (this_week, this_month)",
    "message": "string"
  }
}
```

11.3 Completion Progress (Pillar 6: Behavioral Nudges — Zeigarnik effect)

```
{
  "CompletionProgress": {
    "entity_type": "string (Strategy, FMEA, WorkPackage)",
    "entity_id": "string",
    "entity_name": "string",
    "total_steps": "int (≥1)",
    "completed_steps": "int (≤ total_steps)",
    "completion_pct": "float (0-100, auto-calculated)",
    "remaining_items": ["string"],
    "next_action": "string"
  }
}
```

12. Changelog

Date	Change	Author
2026-02-20	Initial creation — awaiting Discovery	System Pilot
2026-02-20	Discovery Questions answered. Schemas defined. Blueprint phase.	System Pilot
2026-02-20	Module 4 added: Maintenance Strategy Development (full RCM flow + libraries + IA auto-completador). 10 new schemas added.	System Pilot
2026-02-20	GECAMIN extensions: 6 new engines (HealthScore, KPI, Weibull, Variance, CAPA, ManagementReview) + 3 Neuro-Architecture models (ExpertCard, IpsativeFeedback, CompletionProgress). Total: 15 engines, 30+ schemas, 428 tests.	System
2026-02-20	Multi-agent architecture (§7): 4 agents (Orchestrator, Reliability, Planning, Spare Parts), 62 MCP tool wrappers, 4-milestone workflow with human gates, orchestration layer. New tests: 104 (total: 532).	System
2026-02-21	GFSN methodology integration: 5 reference documents (REF-13 to REF-17), 20-gap analysis, Phase 4A engines (RCA, planning KPI, DE KPI, FMECA, GFSN criticality/priority modes), Phase 4B engines (scheduling, execution tasks, work package assembly), Phase 5 engines (jackknife, pareto, LCC, RBI, OCR, MoC, shutdown, spare parts). Full API backend (17 routers, 21 services, 28+ DB models), Streamlit UI (16 pages). Total: 36 engines, 1071 tests.	System
2026-02-22	Phase 8 completion: RCA/DE full stack (3 DB models, API service+router, Streamlit page with 4 tabs, 12 API client functions). Planning KPI snapshots and DE KPI snapshots with history tracking. 14 new API tests. Total: 17 API routers, 17 Streamlit pages, 1085 tests.	System